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Enhanced Interfacial Adhesion and Thermal Stability in Bismuth Telluride/Nickel/Copper Multilayer Films with Low Electrical Contact Resistance

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The interfacial property between thermoelectric films and metal electrodes greatly affects the performance and practical application of thin-film thermoelectric devices. Here, Ni intermediate layer is chosen and inserted into Bi_2Te_3/Cu to simultaneously regulate the electrical and mechanical performance of the interface. Meanwhile, Ar/H_2 plasma cleaning is also adopted to optimize the interfacial connection during the sputtering process. Results show the interfacial element diffusion can be effectively blocked after the introduction of Ni interlayer, and the Bi_2Te_3/Ni/Cu multilayer thin film performs an excellent interfacial adhesion and maintains a low specific contact resistivity about 2.7×10^{-6} Ω cm^2, which can be attributed to the matching of thermal expansion coefficient between Bi_2Te_3, Ni, and Cu, and the reduction of lattice mismatch. Furthermore, the Ni interlayer can also alleviate the degradation of interfacial mechanical and electrical properties after 1000 cycles of thermal shock, consequently enhancing the reliability and stability of the interfacial connection. This work provides an effective way to improve the comprehensive performance of multilayer interface in thin-film thermoelectric devices.

1. Introduction

Thermoelectric (TE) device, as a clean and efficient energy converter, can realize the mutual transformation between thermal energy and electricity over a wide temperature gradient, which is considered as potential candidate for energy conversion.[1–4] For the past few years, studies of thin-film TE devices have become one of the research emphases to enhance the device performance due to its ultrahigh theoretical cooling power density and fast responsivity.[5–8] With decreasing the feature size, interfacial performance between thermoelectric films and metal electrodes has become more crucial on the practical application of thin-film TE device than bulk device,[9] which includes the adhesiveness in interface, the metal–semiconductor (M–S) contact resistance, and thermal stability of interfacial atoms.[10,11]

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In general, the contact resistance can directly affect the device performance.[12] Meanwhile, serious atoms diffusion in the interface would also reduce the service life of thin-film TE devices,[13] and blocking the interface element diffusion excessively may weaken the interfacial bonding force. Hence, how to solve these three problems simultaneously is an enormous challenge. To achieve the required interfacial binding force, introduction of an adhesion layer between the metal and the semiconductor is often employed. For instance, titanium[14] and chromium[15] were employed to enhance the interfacial adhesion strength between metal thin films and ceramic substrates. Furthermore, element diffusion occurred at the interface is also a serious problem that degrades performance of thermoelectric devices in practical application,[16] so a barrier layer needs to be inserted between the metal material and the TE thin film to enhance the reliability and stability of TE devices. Some refractory metals, such as W, Ge, Ti, Zr and their nitrides or alloys, are commonly considered as excellent barrier materials to block atomic diffusion because of their high melting point and chemical stability.[17–20] Thus, choosing an appropriate intermediate layer is the key to optimize the interfacial adhesion and element diffusion collaboratively. Besides, appropriate surface treatment to obtain an ultraclean interface is also necessary to obtain excellent adhesiveness,[21] so a method using Ar/H_2 plasma cleaning was also adopted in this work to optimize the interfacial connection.

It is noticeable that the M–S contact in thermoelectric devices refers to the electrical connection between TE materials and metal electrodes. As reported, surface treatment,[12] ion implantation,[22,23] and multilayer system design of thin films[24,25] have been applied to optimize the contact resistance. In our previous work, a preliminary investigation was also carried out to lower the contact resistance by interfacial structure regulation and a specific contact resistivity of 6.11×10^{-6} Ω cm^2 for Bi_2Te_3/Ti/Cu multilayer films has been obtained.[26] However, researches of interfacial mechanical properties and the degradation of interfacial connection performance under thermal shock are not considered, which also need deeper studies. In addition, the thermal expansion coefficient of intermediate layer at interface plays a vital role in the reliability and stability of the multilayer thin films connection. The linear thermal expansion coefficient

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of Cu, Ni, and Bi_2Te_3 at room temperature are 16.7×10^{-6}, 12.7×10^{-6}, and 12.9×10^{-6} K^{-1}, respectively,[27,28] which indicates that the combination of these three materials is less affected by the temperature change. Meanwhile, the reaction rate of Ni on molten eutectic SnPb solder is about 100 times lower than that of Cu,[29] which makes the solder reaction easier to control at the required degree during the TE device fabrication. Therefore, introducing Ni layer between the TE film and the electrode film is expected to improve the stability and reliability of M–S connection performance. However, the behavior of Ni layer to prevent element diffusion and its specific effects on the electrical and mechanical property of the interface are still unclear, which are still worth further exploring.

Hence in this paper, efforts have been made to insert Ni layer between the thermoelectric thin film (Bi_2Te_3) and the metal electrode film (Cu), and Ar-H_2 plasma cleaning was also introduced to optimize the interfacial connection. Then the interfacial contact performance between Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu was compared and discussed. Besides the M–S electrical contact resistance, interfacial element diffusion, and mechanical adhesion were also considered as important indicators of contact performance. Furthermore, thermal shock test was also conducted to evaluate the reliability and stability of the M–S multilayers.

2. Results and Discussion

2.1. Microstructure and Composition of Multilayer Films

Bi_2Te_3, Bi_2Te_3/Cu, Bi_2Te_3/Ni, and Bi_2Te_3/Ni/Cu multilayer thin films were successively deposited by magnetron sputtering and the crystal structure was investigated by X-ray diffraction (XRD) testing, as shown in **Figure 1**. It can be seen that all of the diffraction peaks of Bi_2Te_3 can be indexed to JCPDS 08-0027 of Bi_2Te_3. The conspicuous diffraction peak

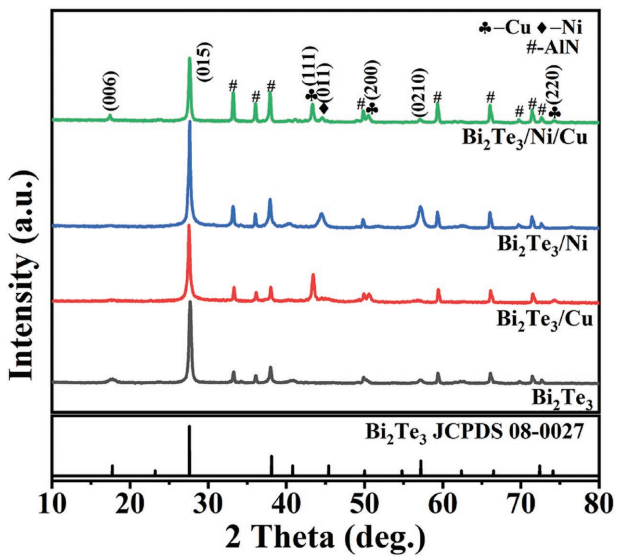


Figure 1. XRD patterns of Bi_2Te_3, Bi_2Te_3/Cu, Bi_2Te_3/Ni, and Bi_2Te_3/Ni/Cu multilayer thin films.

along (015) plane for Bi_2Te_3 in XRD pattern indicates a preferential orientation along (015) plane, and a small account of peaks along (006) or (0210) planes are also detected in Bi_2Te_3 layer. The Cu film exhibits a preferential orientation along (111) plane, and there is no change in diffraction peaks position and intensity of Cu between Cu/Bi_2Te_3 and Cu/Ni/Bi_2Te_3, which indicates that the internal crystal structure of Cu is not affected by Bi_2Te_3 or Ni surface. Diffractograms from Ni/ Bi_2Te_3 and Cu/Ni/Bi_2Te_3 show a single Bragg reflection peak at the position of $2\theta = 44.5°$, indicating a highly (011) oriented structure of Ni crystal. These results clearly show that compound-free samples of nickel layer, copper layer, and Bi_2Te_3 layer have been well fabricated through magnetron sputtering.

Scanning electron microscope (SEM) images in **Figure 2** show surface morphologies and the cross-sectional views of Bi_2Te_3/Cu bilayer, Bi_2Te_3/Ni bilayer, and Bi_2Te_3/Ni/Cu multilayer. In this work, the thickness of Bi_2Te_3 layer in each sample is controlled at about 2 μm, and Cu layer has a thickness of 1 μm. It can be seen that the interface between different materials in each sample can be easily distinguished from their grain features. In Bi_2Te_3/Cu sample, Cu appears as crown-shaped clusters composed of nanoparticles, and the surface of Cu contains a lot of nanocracks. The interface of Bi_2Te_3/Cu presents a point-to-face contact, indicating a poor interfacial contact between Cu and Bi_2Te_3.[30] In Bi_2Te_3/Ni sample, Ni layer displays an obvious compact columnar crystal structure of 700 nm thickness. Bi_2Te_3/Ni sample has a flatter and denser surface topography with fewer void defect in the interface compared to Bi_2Te_3/Cu sample, which is beneficial for improving electrical transport and bonding force. In Bi_2Te_3/Ni/Cu multilayer sample, the Ni layer with thickness of 120 nm is inserted between the Bi_2Te_3 layer and the Cu layer. Figure 2c,f illustrates a significant reduction in surface holes and interfacial defects, and the interface tends to be a face-to-face contact, which indicates that enhanced interface combination and dense Cu layer structure are obtained by introducing Ni layer.

Figure 3 shows element line profiles of Cu/Bi_2Te_3 and Cu/Ni/Bi_2Te_3 samples, where the element distribution of these samples in cross-sectional view is clearly demonstrated. The intensity of Cu in the position of Bi_2Te_3 layer marked by red box indicates that Cu atoms have diffused into Bi_2Te_3 layer during the fabrication process. However, Figure 3b illustrates that there is no obvious Cu signal displayed in the Bi_2Te_3 layer area. The Bi and Te atoms also stay within their region, which proves that Ni can prevent the element diffusion between Cu and Bi_2Te_3 during the thin film preparation. However, Ni atoms have partially diffused into both Bi_2Te_3 layer and Cu layer. Due to the high temperature during sample preparation, atomic diffusion cannot be completely avoided, but Ni only shows slight diffusion compared to Cu, and no further diffusion occurs during the subsequent thermal shock test. Moreover, the steps of sputtering Ni and Cu were carried out continuously in multitarget sputtering system, which causes a period of Ni-Cu cogrowth. The slight diffusion between these two metals can enhance the interfacial adhesion and has no effect on the effectiveness of Ni as a diffusion barrier between Cu and Bi_2Te_3.

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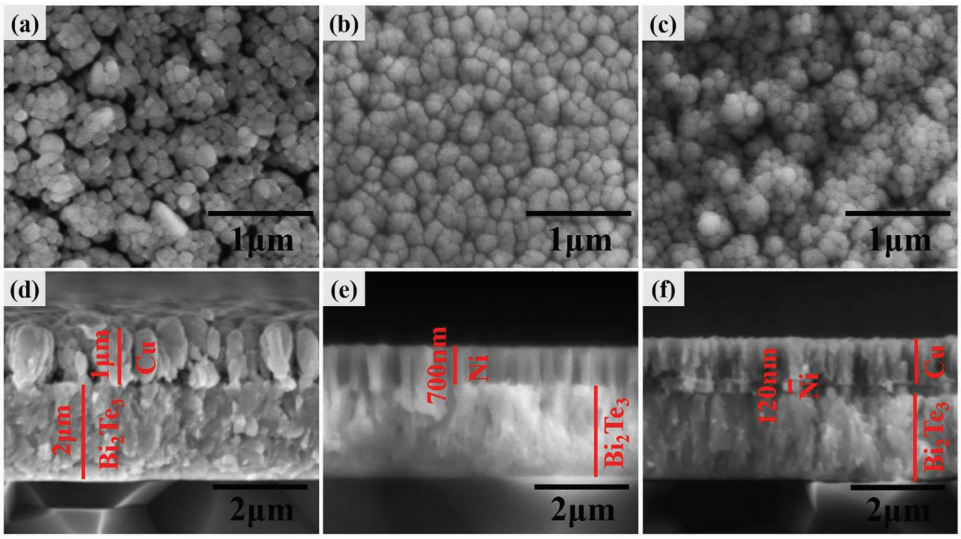


Figure 2. Surface and cross-sectional view of SEM images: a,d) Bi_2Te_3/Cu, b,e) Bi_2Te_3/Ni, and c,f) Bi_2Te_3/Ni/Cu.

2.2. Interfacial Mechanical and Electrical Transport Properties of Multilayer Films

2.2.1. Mechanical Performance of Interfacial Connection

The thin-film TE device usually consists of several layers, including thermoelectric layer, metal electrode layer, solder layer, and various interfacial layer, which are generally connected via physical and/or chemical ways.[31] Therefore, delamination of these heterogeneous materials during manufacture and service has caused widespread concern for device reliability.[32] To enhance the adhesion between TE thin films and metal films, a plasma cleaning process under Ar-H_2 mixture atmosphere was adopted before metal films deposition process. The mechanical bonding performance of multilayer thin films was investigated by nanoscratch test as shown in **Figure 4**. During the test process, the needle tip moves in a specific direction with a constant speed, the force applied on the needle tip increases linearly and eventually the multilayer films are broken. Thereby, the relationship between the scratch depth-scratch distance–tip load is obtained. In this work, the load was set to 0–100 mN with 5 mN s^{-1} of loading rate, and the scratch distance was set to 240 μm with 10 μm s^{-1} of scan velocity. During the scratching process, the load applied on the probe will remain constant for the first 2 s and the last 2 s. As shown in these diagrams, each chart contains three different curves with different colors and these three scratches occur in one position. The prescan curve in green represents the first scratch used for surface detection, and the force applied to the probe keeps constant at 0.1 mN. The scratching curve in red is the second move of tip that breaks into the thin film. The post-scan curve in blue is the third scratch, which is also used for surface detection and the load keeps unchanged with 0.1 mN. This curve contains information on the residual deformation after films break. The obvious undulation on the prescan curves represents large particles of contaminants or pits. The probe first slips through the upper metal films and then into the TE material film during the scratching test. Due to the good ductility of copper, these curves are relatively smooth when the needle passes through the first layer. However, the Bi_2Te_3 layer is hard and brittle, so obvious fluctuation appears in the curve when the needle scratches into the TE material layer, and the sharp drop in depth represents that the Bi_2Te_3 thin films crack in this position. The critical load value of each curve is pointed out, which refers to the corresponding load value when the probe scratches from the ductile metal film into the hard and brittle Bi_2Te_3 film.[33]

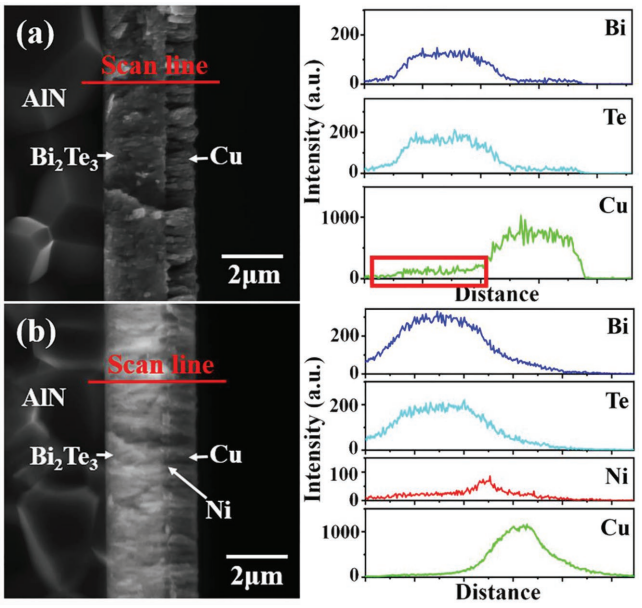


Figure 3. Cross-sectional images and elemental line profiles: a) Bi_2Te_3/Cu and b) Bi_2Te_3/Ni/Cu.

Figure 4a,b represents the nanoscratch results of Bi_2Te_3/Cu sample with and without plasma cleaning process, respectively. A critical load value of 44.25 mN is obtained in Figure 4a and the sample cracks when the scratch depth reaches about 1.8 μm. Figure 4b shows that a higher critical value of

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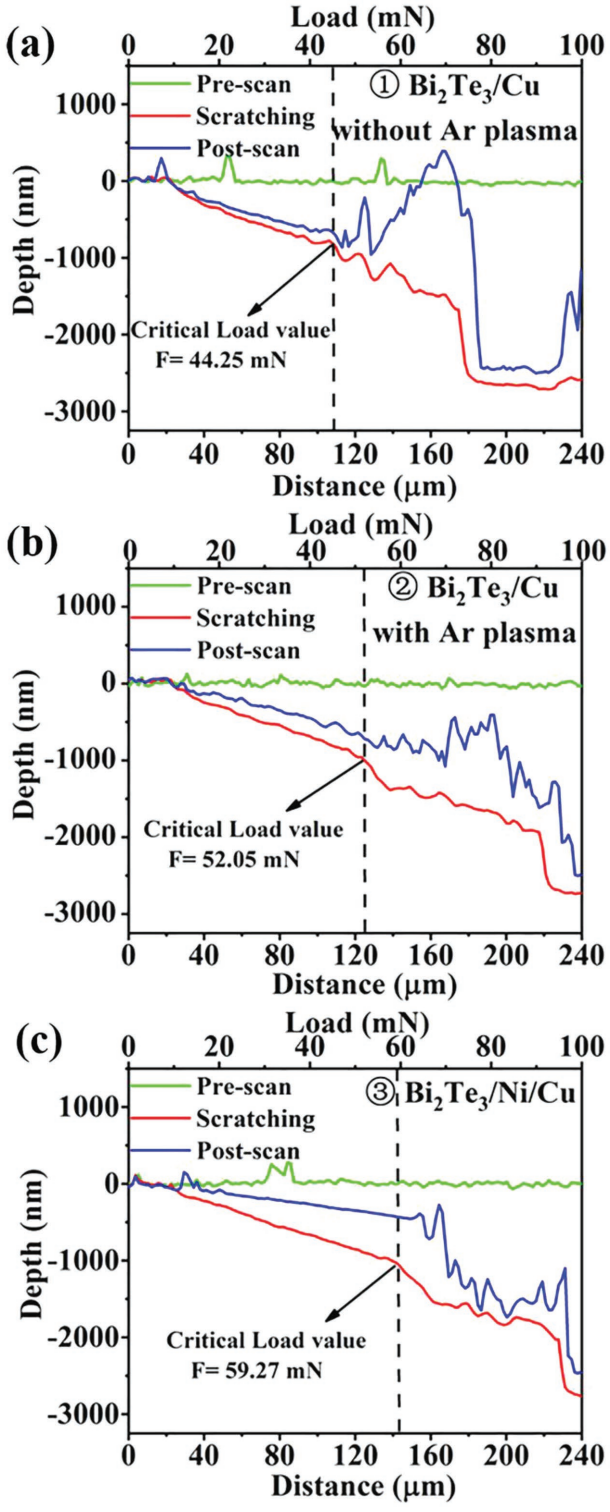


Figure 4. Load-depth and scratch distance curves: a) Bi_2Te_3/Cu, b) Bi_2Te_3/Cu with plasma cleaning, and c) Bi_2Te_3/Ni/Cu with plasma cleaning.

52.05 mN is detected at the same scratch depth. Moreover, the position where films crack in Figure 4b is further back compared with that in Figure 4a. Obviously, the plasma cleaning process has enhanced interfacial bonding, which may be attributed to

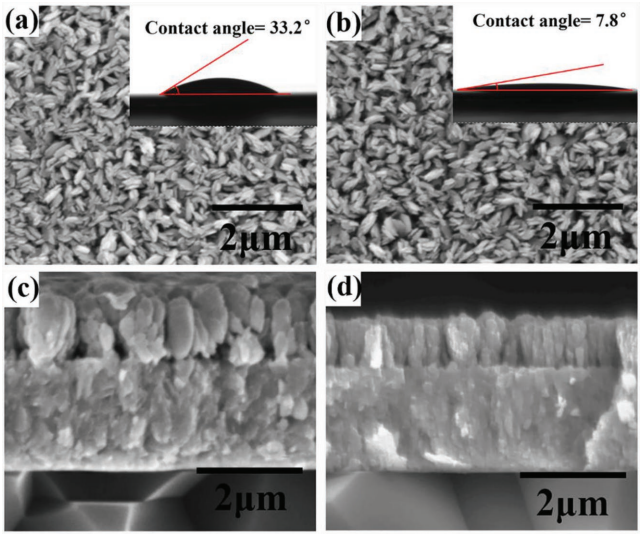


Figure 5. Surface morphology of SEM images of Bi_2Te_3 and cross-sectional view of SEM images of Bi_2Te_3/Cu: a,c) without plasma cleaning and b,d) with plasma cleaning. Insets are contact angle pictures of related surface.

the optimization of interfacial wettability and surface energy. In addition, after the introduction of Ni layer, the critical load value is further increased to 59.27 mN in Figure 4c, indicating a favorable junction in the interface of Bi_2Te_3/Ni/Cu films.

The contact angle measurement was adopted to investigate the change in wettability and surface energy of the Bi_2Te_3 surface after plasma cleaning. As shown in **Figure 5**, after the Ar-H_2 plasma cleaning, the contact angle decreases from 33.2° to 7.8°, and the calculated surface energy increases from 67.8 to 75 mJ m^{-2}. Comparing the cross-sectional view in Figure 5c,d, the grain size of Cu decreases and the interfacial connection becomes compact after Ar-H_2 plasma cleaning. These results can be attributed to the improvement of wettability and surface energy of Bi_2Te_3, which increases the nucleation rate of Cu on the Bi_2Te_3 surface and reduces the interfacial porosity, thus improving the adhesion between Bi_2Te_3 and Cu. That is also the reason why the critical load value has been enhanced after Ar-H_2 plasma cleaning in Figure 4b. In addition, it has been reported that the lattice mismatch factor (F) between hexagonal Ni and hexagonal Bi_2Te_3 based material ($Bi_2Te_{2.7}Se_{0.3}$) is far smaller than that between cubic Cu and hexagonal BTS.[34] According to the XRD result in Figure 1, lattice parameters are $a = b = 0.430$ nm and $c = 2.991$ nm for hexagonal Bi_2Te_3, $a = b = 0.2665$ nm and $c = 0.4343$ nm for hexagonal Ni, and $a = b = c = 0.3615$ nm for cubic Cu. Thus, the calculated F value is about 19% for Bi_2Te_3/Cu and about 7% for Bi_2Te_3/Ni, which is well matched with above results and leads to a further strengthening of interfacial bonding.

2.2.2. Electrical Contact Resistance

When metal and semiconductor are connected, a potential barrier may form at the interface due to the mismatch of the work function between the metal and the semiconductor, which is

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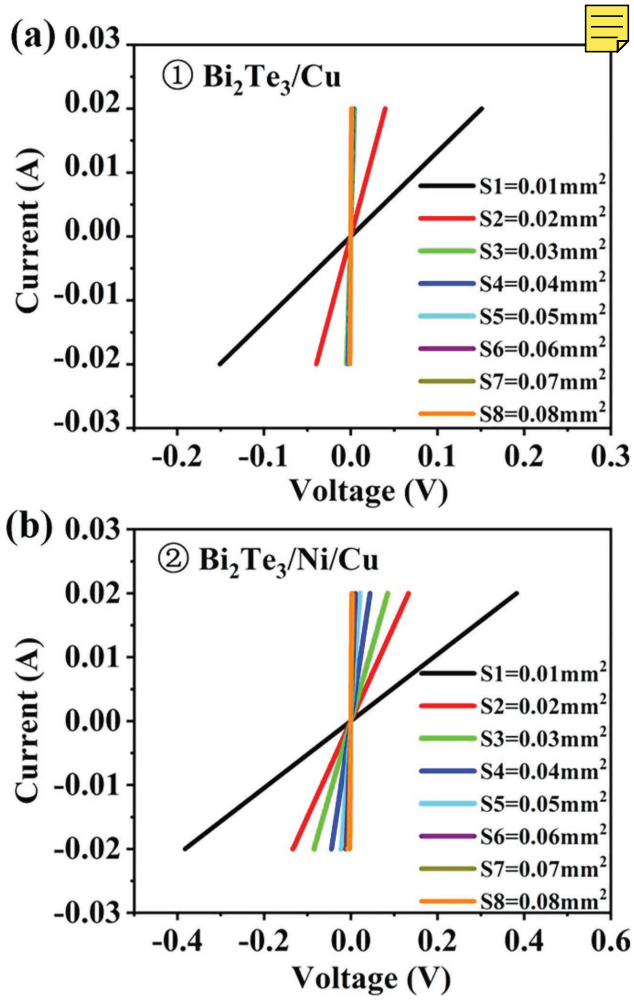


Figure 6. *I–V* test results: a) Bi_2Te_3/Cu and b) Bi_2Te_3/Ni/Cu.

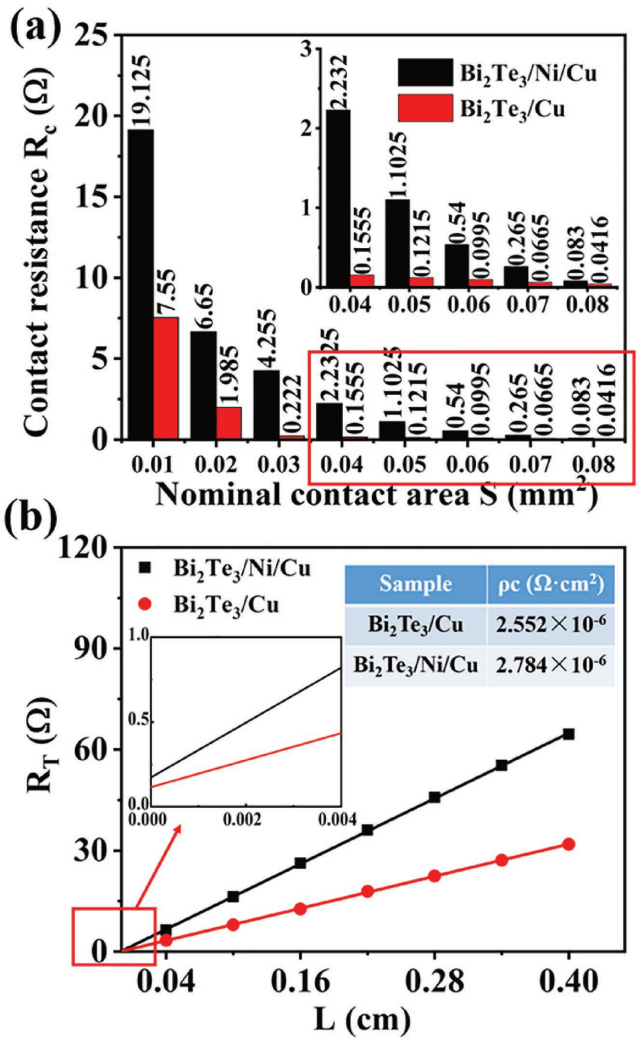


Figure 7. a) Contact resistance histogram of Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu at different contact areas. b) Total resistance (R_T) versus metal pad spacing (L) plot of Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu with ρ_C extracted in the table.

called Schottky barrier.[25] The Schottky barrier can block the current flow in one direction and results in high contact resistance at the M–S interface. Therefore, avoiding the formation of Schottky barrier is a precondition for a good M–S electrical contact. As shown in **Figure 6**, the three-probe method with *I–V* test was applied to investigate the contact in the M–S junction. The excellent symmetrical linear relationship illustrates the formation of Ohmic contact.

Through further analyzing these data, the contact resistance at each contact area can be extracted from the slope of each line. **Figure 7**a shows the contact resistance in different contact areas, where the variation meets with our previous work that the contact resistance decreases nonlinearly with the increase of contact area.[26] At the interface between the metal and the semiconductor, the contact resistance (R_c (Ω)) changes with the contact area (S (cm^2)), but the specific contact resistivity (ρ_C (Ω cm^2)) is constant. The relationship between R_c and ρ_c is $R_C = \rho_C / S$. Thus, the contact resistance decreases as the contact area increases. It can be seen that the contact resistance of Bi_2Te_3/Ni/Cu is larger than that of the Bi_2Te_3/Cu at each specific contact area. When the contact area exceeds 0.03 mm^2, the R_C of Bi_2Te_3/Ni/Cu continuously decreases with the increase of S, and the R_C of Bi_2Te_3/Cu remains substantially stable, which indicates that the R_C of Bi_2Te_3/Ni/Cu is more sensitive to contact area variation. These results are critical for dimensional design in the miniaturization technology for thin-film TE device.

Remarkably, the contact area in our paper refers to the designed nominal contact area, and the actual contact area in the interface will deviate from the designed size due to the surface roughness and the interfacial crystal structure.[35] Thus, the transmission line mode (TLM) method has been adopted to investigate the specific contact resistivity, which is shown in Figure 7b. Data points in the graph represent the total resistance (R_T) between two copper electrodes pad at different spacing (L), and each R_T is also obtained from 2000 date points in *I–V* test. It can be seen that there is a good linear relationship between these data points in each sample, ρ_C of Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu were extracted and listed in the table beyond these fitted lines. The electrical transport of Bi_2Te_3/Cu sample exhibits a low specific contact resistivity of 2.552×10^{-6} Ω cm^2,

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and the ρ_C is slightly increased to 2.784×10^{-6} Ω cm^2 after the introduction of Ni interlayer. Practically, the contact resistance between the metal and the semiconductor is affected by many factors including work function, surface state, defects and impurities, etc.[36] On the one hand, the optimized interfacial combination after Ni introduction improves the interfacial electrical transport. On the other hand, the work function difference between Ni (4.6 eV) and Bi_2Te_3 (5.3 eV) is larger than that of Cu (4.65 eV) and Bi_2Te_3.[22] Therefore, the ρ_c of Bi_2Te_3/Ni/Cu sample is slightly higher than that of Bi_2Te_3/Cu, but which is also controlled at a very low level.

2.3. Interfacial Stability of Multilayer Films after Thermal Shock

2.3.1. Interfacial Element Diffusion after Thermal Shock

Figure 8 shows the composition depth profiles of Cu/Bi_2Te_3 and Cu/Ni/Bi_2Te_3 after thermal cycling. Green lines of Cu signal show visible difference between Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu. As the cycle times increase, the Cu signal intensity in Bi_2Te_3 layer region is gradually enhanced, indicating that Cu atoms increasingly penetrate into Bi_2Te_3 layer. After the introduction of Ni interlayer, significant Ni signal (red line) can be detected at the position between Cu and Bi_2Te_3 and the Cu signal on Bi_2Te_3 layer can be basically kept at a very low intensity during the thermal shock process, representing that Ni interlayer can play a good role as a barrier to block atoms diffusion.

Atomic diffusion can be divided into these following paths: crystal lattice diffusion (interstitial diffusion, vacancy diffusion, and substitution diffusion), grain boundaries diffusion, dislocation diffusion, and surface diffusion.[16] The crystal lattice diffusion generally occurs at high temperature because atoms on the lattice node are relatively stable. Atoms near the grain boundary and dislocation have higher energy and easily diffuse along these defects, the interface is where atoms diffusion is most likely to occur due to the smallest diffusion activation energy. According to the temperature in practical application and the thermal cycling, these elements diffusion mainly occurs by last three ways. As we mentioned above, the large lattice mismatch between Cu and Bi_2Te_3 causes a lot of defects and voids existing in the interface, and this unstable contact can easily produce more defects and microcracks during the thermal cycling progress. Hence, the presence of defects accelerates the atomic diffusion, and thus lowers the reliability of multilayer films.

2.3.2. Mechanical Properties after Thermal Shock

Figure 9a1,2 illustrates the nanoscratch test results of Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu samples after 1000 cycles of thermal shock. The critical load value decreases from 52.04 to 42.04 mN and from 59.27 to 51.47 mN for the Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu samples, respectively. Furthermore, there is a sharp drop in the scratching curve at the critical load location, which indicates that the underlying film has cracked when the needle reaches the bismuth telluride. Since Bi_2Te_3/Ni/Cu and Bi_2Te_3/Cu films were both seriously degraded in interfacial adhesion after 1000 cycles of thermal shock, films after 400 cycles of thermal shock were selected to compare their morphology changes. Figure 9a2 clearly shows the continuous cracks appearing on the surface of Bi_2Te_3/Cu films after 400 cycles of thermal shock, and obvious peeling also appears in Figure 9a3. However, Bi_2Te_3/Ni/Cu films show no obvious cracks after 400 cycles of thermal shock, which demonstrates a preferable interfacial connection. Therefore, the degradation of mechanical performance can be attributed to two factors: (i) the internal stress accumulation of thin films is produced by the thermal shock, when the needle reaches the hard and brittle Bi_2Te_3 thin film, and stress release causes films to break; (ii) because of the difference of thermal expansion coefficient, microcracks and defects extend at the

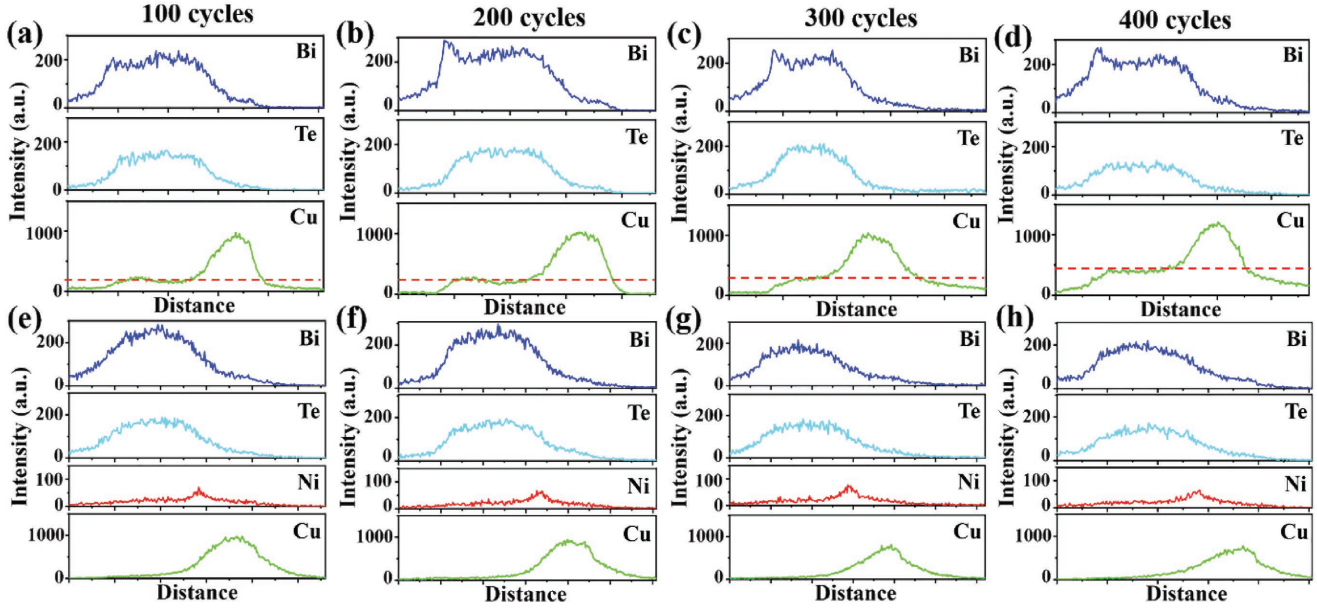


Figure 8. Elemental line profiles after different cycles of thermal shock: a–d) Bi_2Te_3/Cu and e–h) Bi_2Te_3/Ni/Cu.

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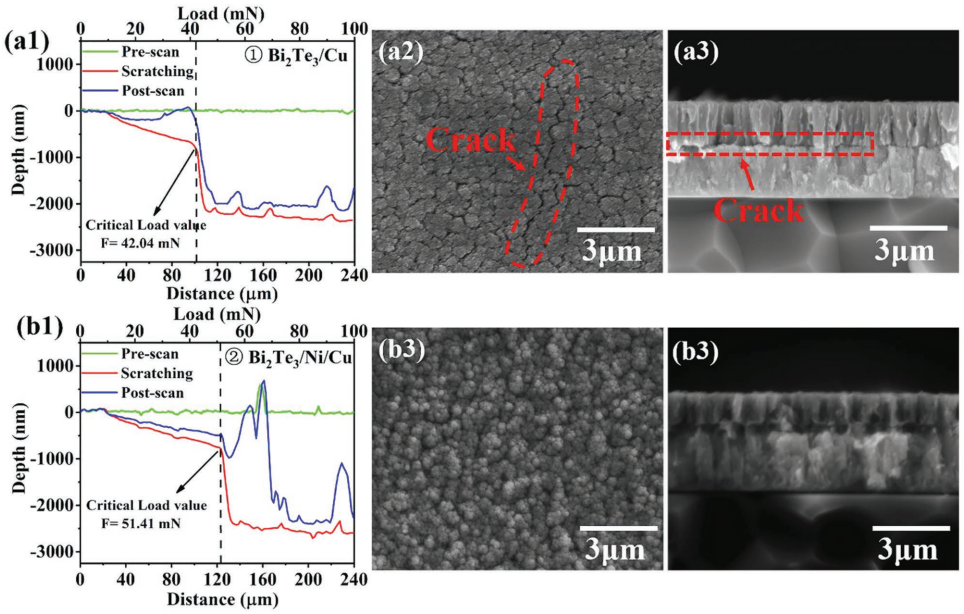
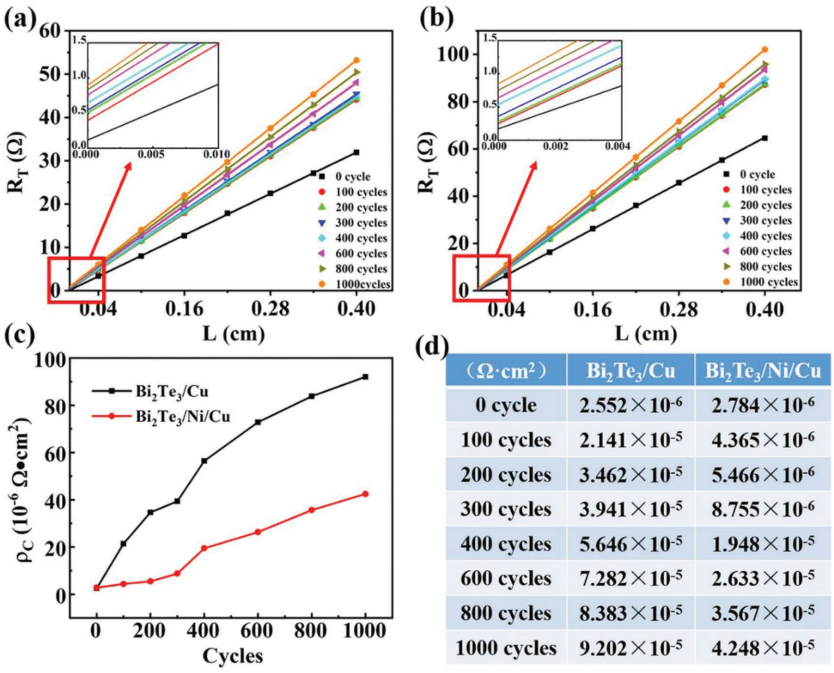


Figure 9. Load-scratch depth-scratch distance curves after 1000 cycles of thermal shock, Surface and cross-sectional view of SEM images after 400 cycles of thermal shock. Cracks are marked by red dashed lines. a) Bi_2Te_3/Cu and b) Bi_2Te_3/Ni/Cu.

interface under the thermal shock. However, the Bi_2Te_3/Ni/Cu still exhibits a higher critical load value than that of the Bi_2Te_3/Cu because of the improvement of interfacial connection.

2.3.3. Interfacial Contact Resistance after Thermal Shock

Figure 10 shows the TLM method test result of Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu samples after thermal shock, and lines in Figure 10a,b with different colors represent test results after different cycles of thermal shock. It can be seen that the R_T of each data point increases with the thermal shock cycles, resulting in a rise of both the slope and the *Y*-axis intercept of their lines. A line chart of ρ_C is shown in Figure 10c. It is obvious that the ρ_C of both Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu increases with the thermal shock cycles. Remarkably, there is a severe degradation of Bi_2Te_3/Cu after 100 cycles of thermal shock, and the ρ_C increased by one order of magnitude. However, the ρ_C of Bi_2Te_3/Ni/Cu sample degenerates more slowly compared to Bi_2Te_3/Cu sample. In short, the specific contact resistivity is increased from 2.552×10^{-6} to 9.202×10^{-5} Ω cm^2 for Bi_2Te_3/Cu contact and from 2.784×10^{-6} to 4.248×10^{-5} Ω cm^2 for Bi_2Te_3/Ni/Cu contact after 1000 cycles of thermal shock. The contact resistance between the metal and the semiconductor is influenced by several factors including work function, surface state, defects and impurities, second phases, grain size and orientation.[36] These factors affect the barrier height and/or width, which consequently affect the M–S contact resistance. As we mentioned above, Figure 8 shows apparent element diffusion in the Bi_2Te_3/Cu during the thermal shock process. Actually, it has been reported that the suppression of some compounds formed at the interface between Cu and n-type Bi_2Te_3 favors interfacial electrical transport.[13] However, the



(Ω·cm^2)	Bi_2Te_3/Cu	Bi_2Te_3/Ni/Cu
0 cycle	2.552×10^{-6}	2.784×10^{-6}
100 cycles	2.141×10^{-5}	4.365×10^{-6}
200 cycles	3.462×10^{-5}	5.466×10^{-6}
300 cycles	3.941×10^{-5}	8.755×10^{-6}
400 cycles	5.646×10^{-5}	1.948×10^{-5}
600 cycles	7.282×10^{-5}	2.633×10^{-5}
800 cycles	8.383×10^{-5}	3.567×10^{-5}
1000 cycles	9.202×10^{-5}	4.248×10^{-5}

Figure 10. TLM test results after different cycles of thermal shock. a) Bi_2Te_3/Cu; b) Bi_2Te_3/Ni/Cu; c) comparison for specific contact resistivity changes between Bi_2Te_3/Cu and Bi_2Te_3/Ni/Cu; d) the specific value of ρ_C.

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interfacial problem is commonly complicated. The diffusion of Cu into Bi_2Te_3 may improve the electrical conductivity of Bi_2Te_3 by doping and consequently enhance the electrical transport properties at the interface.[37] Besides, Figure 9 shows Bi_2Te_3/Ni/Cu and Bi_2Te_3/Cu films are both degraded in interfacial adhesion after 1000 cycles of thermal shock, but the degradation of Bi_2Te_3/Cu films is more serious than that of Bi_2Te_3/Ni/Cu. The SEM photos in Figure 9 also illustrate that Bi_2Te_3/Cu films produce significant cracks after 400 cycles of thermal shock. Thus, in this work, the degeneration in electrical transport properties is mainly attributed to the interfacial element diffusion and the weakening of mechanical performance. Compared to Bi_2Te_3/Cu sample, the Ni interlayer in Bi_2Te_3/Ni/Cu can effectively block the element diffusion in the interface and it also can well connect to both Bi_2Te_3 layer and Cu layer, therefore improving the stability and reliability of the multilayer thin films.

3. Conclusion

Generally, Bi_2Te_3/Ni/Cu multilayer thin films have been fabricated by magnetron sputtering technology. The Ni interlayer is proved to be an effective barrier layer to block the interfacial element diffusion even after 1000 cycles of thermal shock. Meanwhile, the plasma cleaning is also adopted before metal films deposition to enhance the interfacial bonding force. Nanoscratch test results show that the critical load has been improved from 44.25 to 52.05 mN after plasma cleaning in Bi_2Te_3/Cu sample, which is further improved to 59.27 mN after the introduction of Ni interlayer. A low specific contact resistivity about 2.7×10^{-6} Ω cm^2 is obtained for Bi_2Te_3/Ni/Cu, which is attributed to the matching of thermal expansion coefficient between Bi_2Te_3, Ni, and Cu, and the reduction of lattice mismatch. Furthermore, the contact resistivity is increased from 2.552×10^{-6} to 9.202×10^{-5} Ω cm^2 for Bi_2Te_3/Cu contact and from 2.784×10^{-6} to 4.248×10^{-5} Ω cm^2 for Bi_2Te_3/Ni/Cu contact after 1000 cycles of thermal shock, strongly illustrating that the Ni interlayer has improved the stability and reliability in the M–S contact between Cu and Bi_2Te_3. This study should be helpful to improve the stability of Bi_2Te_3-based thin-film thermoelectric devices. Besides, this method could also be extended to other thin-film metal electrodes like Ni/Ag and Ni/Al or p-type thermoelectric films.

4. Experimental Section

Preparation of Thin Film and Patterning: In this work, multilayer thin film samples were deposited on insulated fine polished aluminum nitride (AlN 42 mm × 33 mm) by magnetron sputtering technology under Ar atmosphere using a magnetron sputtering system (JGP-450a, SKY Technology Development Co., Ltd., Chinese Academy of Sciences). Before deposition, the AlN substrate underwent a 15 min ultrasonic bath cleaning in detergent, deionized water, acetone, and alcohol, respectively. The distance between the AlN substrate and the target was 80 mm and the substrate kept a 5 turns min^{-1} rotation during the sputtering process to guarantee a homogeneous deposition. Mask plates were used to obtain the contact resistance testing pattern. Detailed experimental parameters used for the sputtering are described in **Table 1**. A 120 s plasma cleaning process (PT-05-LF, Tiankechuangda Co., Ltd.) under Ar-H_2 atmosphere was applied to enhance the interfacial connection before films deposition. After the preparation of thin films, a thermal shock test was conducted (CJX, Taixing Weida Test Equipment Co., Ltd., China), and the temperature was set to −40 °C in the cold-end and 120 °C in the hot-end with each duration time of 2 min.

Table 1. Specific experimental parameters used in magnetron sputtering.

Layer	Target (Φ 60 mm, 99.99%)	Power [W]	Substrate temperature [°C]	Working pressure [Pa]	Sputtering time [min]
Bi_2Te_3	Bi_2Te_3	DC[a] 20	200	2	90
	Te	RF[b] 25			
Ni	Ni	DC 30	100	1	20
Cu	Cu	DC 30	200	1	120

[a]DC represents direct current power; [b]RF represents radio frequency power.

Characterization and Measurement: The phase composition and crystal structure of films were determined by XRD (Rigaku D/MAX 2200 PC) using the Cu Kα radiation (λ = 0.154 nm). The morphology of thin films was observed on a field emission scanning electron microscope (FESEM) (FEI Sirion 200), and the chemical composition was examined by energy-dispersive X-ray spectroscopy (EDS). The wettability and surface energy of thin films were measured by XG-CAM contact angle meter (Shanghai Xuanyichuangxi Industrial Equipment Co.,Ltd.). Nanoscratch test (Nano Test P3, Micro Materials Ltd., Wrexham, UK) was adopted to investigate the mechanical properties of films junction. The electrical contact property was characterized by the TLM method and three-probe method using a ST-102A probe testing platform equipped with a Keithley 2400 digital source meter.

Electrical Contact Properties Investigation: The electrical transport property was investigated by three-probe method and TLM method according to a previous work.[26] As shown in **Figure 11**a, different contact areas from 0.1 to 0.8 mm^2 between metal pads and Bi_2Te_3 in three-probe method were designed. The contact resistance is derived from the *I*–*V* test which is operated on the probe testing platform. Current scan mode (sweeping from −20 to 20 mA) was selected to get the *I*–*V* curve, and 2000 points of voltage were recorded during the process. In TLM method, the total resistance (R_t) between two metal pads consists of the resistance of Bi_2Te_3 between two metal pads and the contact resistance (R_c) of two contact surfaces, which can be described as Equations (1) and (2)[38]

$$R_t = \frac{R_{sh}}{W} L + \frac{2L_T R_{sk}}{W} \quad (1)$$

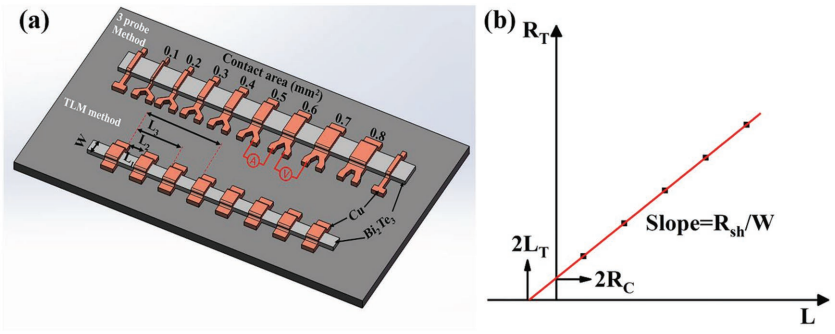


Figure 11. a) Principle diagram for three probe method and TLM method measurement system. b) A diagram of the R_T versus L plot, where ρ_C can be extracted according to the fitted line.

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$$R_c = \frac{L_T R_{sk}}{W} \quad (2)$$

where R_{sh} and R_{sk} are the sheet resistance of Bi_2Te_3 and the modified sheet resistance of Bi_2Te_3 under the contact, respectively; L_T is the transfer length, which is related to the distance that current needs to flow in or out the contact surface; W is the contact width of the overlap area; and L is the distance between two measured metal pads.

Through plotting R_t versus L according to Equation (1), a straight line by linear fitting is illustrated in Figure 11b. The R_c can be obtained from the *y*-intercept, the L_T can be obtained from the *x*-intercept, and the slope represents R_{sh}/W. R_{sh} and R_{sk} are assumed to be equal in this work, so the effective contact area can be expressed with $W \times L_T$. Therefore, the specific contact resistivity is given by Equation (3)

$$\rho_c = R_c \times W \times L_T \quad (3)$$

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrical contact resistivity, element diffusion, interfacial adhesion, thermal stability

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[1] T. Kil, S. Kim, D. Jeong, D. Geum, S. Lee, S. Jung, S. Kim, C. Park, J. Kim, J. M. Baik, K. Lee, C. Z. Kim, W. J. Choi, S. Baek, *Nano Energy* **2017**, *37*, 242.
[2] L. E. Bell, *Science* **2008**, *321*, 1457.
[3] J. A. Paradiso, T. Starner, *IEEE Pervasive Comput.* **2005**, *4*, 18.
[4] K. Biswas, J. He, I. D. Blum, C. Wu, T. P. Hogan, D. N. Seidman, V. P. Dravid, M. G. Kanatzidis, *Nature* **2012**, *489*, 414.
[5] K. Tappura, *Renewable Energy* **2018**, *120*, 78.
[6] N. S. Hudak, G. G. Amatucci, *J. Appl. Phys.* **2008**, *103*, 101301.
[7] R. He, G. Schierning, K. Nielsch, *Adv. Mater. Technol.* **2018**, *3*, 1700256.
[8] Z. Zheng, J. Luo, T. Chen, X. Zhang, G. Liang, P. Fan, *Appl. Phys. Lett.* **2018**, *112*, 163901.
[9] W. Liu, Q. Jie, H. S. Kim, Z. Ren, *Acta Mater.* **2015**, *87*, 357.
[10] L. W. Da Silva, M. Kaviany, *Int. J. Heat Mass Transfer* **2004**, *47*, 2417.
[11] D. Zhao, H. Geng, L. Chen, *Int. J. Appl. Ceram. Technol.* **2012**, *9*, 733.
[12] R. P. Gupta, K. Xiong, J. B. White, K. Cho, H. N. Alshareef, B. E. Gnade, *J. Electrochem. Soc.* **2010**, *157*, H666.
[13] T. Cardinal, Devender, T. Borca-Tasciuc, G. Ramanath, *ACS Appl. Mater. Interfaces* **2016**, *8*, 4275.
[14] V. A. Venugopal, G. Ottaviani, C. Bresolin, D. Erbetta, A. Modelli, E. Varesi, *J. Electron. Mater.* **2009**, *38*, 2063.
[15] C. Yu, C. Chan, L. Chan, K. Hsieh, *Microelectron. Reliab.* **2011**, *51*, 119.
[16] R. Li, B. Qiao, H. Shang, J. Zhang, C. Jiang, W. Zhang, *J. Alloys Compd.* **2018**, *748*, 258.
[17] D. Kim, Y. Jung, M. Lee, Y. Lee, J. Lee, J. Lee, *Thin Solid Films* **2000**, *372*, 276.
[18] S. Chandrasekaran, F. M. Simanjuntak, T. Tseng, *Jpn. J. Appl. Phys.* **2018**, *57*, 04FE10.
[19] J. S. Becker, R. G. Gordon, *Appl. Phys. Lett.* **2003**, *82*, 2239.
[20] S. Rawal, D. P. Norton, K. Kim, T. J. Anderson, L. McElwee-White, *Appl. Phys. Lett.* **2006**, *89*, 231914.
[21] J. D. Lim, P. M. Lee, D. M. W. Rhee, K. C. Leong, Z. Chen, *Appl. Surf. Sci.* **2015**, *355*, 509.
[22] P. J. Taylor, J. R. Maddux, G. Meissner, R. Venkatasubramanian, G. Bulman, J. Pierce, R. Gupta, J. Bierschenk, C. Caylor, J. D. Angelo, Z. Ren, *Appl. Phys. Lett.* **2013**, *103*, 043902.
[23] M. Koike, Y. Kamimuta, E. Kurosawa, T. Tezuka, *Appl. Phys. Express* **2014**, *7*, 051302.
[24] H. Keum, H. Chung, S. Kim, *ACS Appl. Mater. Interfaces* **2013**, *5*, 6061.
[25] K. Byun, H. Chung, J. Lee, H. Yang, H. J. Song, J. Heo, D. H. Seo, S. Park, S. W. Hwang, I. Yoo, K. Kim, *Nano Lett.* **2013**, *13*, 4001.
[26] X. Kong, W. Zhu, L. Cao, Y. Peng, S. Shen, Y. Deng, *ACS Appl. Mater. Interfaces* **2017**, *9*, 25606.
[27] B. Zhang, X. Li, D. Li, *Calphad* **2013**, *43*, 7.
[28] L. M. Pavlova, Y. I. Shtern, R. E. Mironov, *High Temp.* **2011**, *49*, 369.
[29] K. N. Tu, K. Zeng, *Mater. Sci. Eng., R* **2001**, *34*, 1.
[30] D. Jeong, R. Abdur, Y. Joo, J. Jang, P. R. Cha, J. Kim, K. Min, J. Lee, *Mater. Sci. Semicond. Process.* **2018**, *83*, 33.
[31] W. Zhu, Y. Deng, L. Cao, *Nano Energy* **2017**, *34*, 463.
[32] L. Cai, P. Li, P. Wang, Q. Luo, P. Zhai, Q. Zhang, *J. Electron. Mater.* **2018**, *47*, 5.
[33] B. D. Beake, V. M. Vishnyakov, A. J. Harris, *Surf. Coat. Technol.* **2017**, *309*, 671.
[34] H. Zhou, X. Mu, W. Zhao, D. Tang, P. Wei, W. Zhu, X. Nie, Q. Zhang, *Nano Energy* **2017**, *40*, 274.
[35] R. L. Jackson, S. H. Bhavnani, T. P. Ferguson, *J. Heat Transfer* **2008**, *130*, 081301.
[36] M. Muñoz-Rojo, O. Caballero-Calero, M. Martín-González, *Appl. Phys. Lett.* **2013**, *103*, 183905.
[37] W. Liu, Q. Zhang, Y. Lan, S. Chen, X. Yan, Q. Zhang, H. Wang, D. Wang, G. Chen, Z. Ren, *Adv. Energy Mater.* **2011**, *1*, 577.
[38] M. Lijadi, F. Pardo, N. Bardou, J. Pelouard, *Solid-State Electron.* **2005**, *49*, 1655.

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